A complete fourteen-variable consistency equation

Strictly positive integer unknowns; exact degree 150; a 65-operation evaluator.

The only unknowns are

$$J, F, \alpha, z, f, i, T, s, w, n, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

All other symbols in the equation are polynomial abbreviations:

$$\begin{gathered}
q_- = \beta J, \quad q = q_- + 1, \quad C = q - F - \alpha, \\
X = wq, \quad Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \\
\ell = EkY, \quad c = kY + \eta, \quad a = E + Y, \\
H = 4a + 3, \quad \Delta = a^2 + H, \quad D = X + ac + \gamma H, \\
R = [q_-(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
\tau = En + k - R, \quad V = f^2(cT - R) - c, \\
S = i\Delta c^2, \\
N_0 = \tau^2 - \ell(\ell + k), \quad N_1 = D^2 - \Delta c^2, \\
N_a = S^2(V^2 - y^2) + y^2, \\
N_t = (K + w)C + (q - F) - zq_-, \quad N_s = \Delta f^2 - S^2.
\end{gathered}$$

The single equation is

$$\boxed{N_0 N_1 N_a N_t N_s - \Delta = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho = 2^{47}, \quad B = \rho^{4027}, \quad \beta = B - 1, \\
G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1992} + \rho^{2998}.
\end{gathered}$$

$$\begin{aligned}
M_C = {} & \beta - \rho^{977} \\
& - G_{942}(\rho) - \Psi(\rho, 1), \\
M_F = {} & B + 3 + (2^{44} - 2)\Lambda \\
& + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1005} \\
& + 2^{27}\rho^{4022} - \rho^{4004}, \\
K = {} & \rho^{18} + \rho^{1005} + \rho^{4036} \\
& + \Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) = {} & \sum\nolimits_{i=0}^{941} [1 + 2^{10}(1 + w_i + 2^8 n_i \\
& + 2^{17} p_i + 2^{25} s_i)] x^i + 2^{10}\Psi(x, 2), \\
\Psi(x, t) = {} & x^{969}(x^9 + t^{17}) G_8(tx) \\
& + t^8 x^{942}(x^{18} + t^{17}) G_9(tx) \\
& + (2 - t)x^{1005}.
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$\begin{gathered}
h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\}, \\
r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j + 1$ $(j = 2q + b < 430)$	$((1 - d_j)r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$	$(dr_{d,q}, b, (1 - d)r_{d,q}, h(q, b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
N = \mathcal{A} \bmod 2^{636}, \quad Q = \lfloor \mathcal{A}/2^{636} \rfloor, \\
\nu_j = D_2(N, j), \quad \sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \\
r_j = (\sigma_j + 2)^{1 - \nu_j}, \quad P_j = \prod_{k=0} r_k \ (j \le 430), \\
m_q = D_2(N, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14 + m_q(D_{16}(\lfloor Q/P_{430} \rfloor, \tau_q) - 14), \\
u_j = (\sigma_{j+1} - (\lfloor Q/P_j \rfloor \bmod r_j)) \bmod 216, \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \dots, 61$, is `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`. Concatenate the following 399 characters as digits $e_0, \dots, e_{398}$; put $\mathcal{A} = \sum_{j=0}^{398} e_j 62^{398-j}$.

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5TPXxOMcEgSbYki3sGVNKJLlN8Q3rCXEfmrabEMMQ8k2o7MWvDH3pCkVAr7WqE
SjjWJ6dfq6LXnsZ6OHq7X5iVUPDlwMCAVZSQetoQOToXWH6MDnc0OLmAYqNDjl
4vOt5Pyqlm1w3QNJbgusSWi8PpjYYLOGfwYX6CiaBwhYAamgcErqONvxD1RlnG
hk2ZQlIzR5QYJ7qPAGs4LiKjiwLVxBklwY6mcOrXASKtPvKSExJCqmSqOQA4We
dyi3CmNTqaYAAUHXNKViiOY7dPyuBGK3Wf9xU3vNHYczDPSJUREK7toNXVZLEO
Av8LCOsn6Z6b3e6vj2YEsfHM9lAXImUXzNvBajRskYaVHYhxtRPsiISOORst1h
16ymUkkXf1F2lVOfi8rCWgfBlPp
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The ordinary complemented-color, native and Pell arguments give unsolvability iff Con(ZFC). Native calibration and internal-ZFC formalization remain open in Lean. This page is a complete finite specification, not an unsolvability proof.